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Citation: *American Journal of Physics* **82**, 224 (2014); doi: 10.1119/1.4848135

View online: <http://dx.doi.org/10.1119/1.4848135>

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Thermodynamics of bread baking: A two-state model

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(Received 19 March 2013; accepted 21 November 2013)

Bread baking can be viewed as a complex physico-chemical process. It is governed by transport of heat and is accompanied by changes such as gelation of starch, the expansion of air cells within dough, and others. We focus on the thermodynamics of baking and investigate the heat flow through dough and find that the evaporation of excess water in dough is the rate-limiting step. We consider a simplified one-dimensional model of bread, treating the excess water content as a two-state variable that is zero for baked bread and a fixed constant for unbaked dough. We arrive at a system of coupled, nonlinear ordinary differential equations, which are solved using a standard Runge-Kutta integration method. The calculated baking times are consistent with common baking experience. © 2014 American Association of Physics Teachers.
[\[http://dx.doi.org/10.1119/1.4848135\]](http://dx.doi.org/10.1119/1.4848135)

I. INTRODUCTION

The underlying principles of bread baking are explained in the popular literature.^{1,2} The seeming simplicity of baking is deceptive, however, and the details are enormously difficult to model, as is apparent from a cursory look through the standard 1400-page reference.³ Various raw materials (flour, sugar, salt, fat, yeast, and others) have different physico-chemical properties and undergo transformations driven by temperature changes and mass transfer.

Baking involves the flow of heat from the outside (crust) to the center of the bread. The energy current through an area A , $j = A^{-1} \partial Q / \partial t$, is proportional to the temperature gradient: $j = -k \nabla T$, where k is the thermal conductivity.⁴ The time dependence of the temperature in a region with no heat sources or sinks is determined by the diffusion equation⁵ $\nabla^2 T = (c\rho/k) \partial T / \partial t$, where c is the specific heat and ρ is the density. We assume the same values in dough and bread: $c = 2 \times 10^3 \text{ J}/(\text{kg} \cdot \text{K})$, $\rho = 1 \times 10^3 \text{ kg}/\text{m}^3$, and $k = 0.5 \text{ J}/(\text{s} \cdot \text{m} \cdot \text{K})$, so that $c\rho/k = 4 \times 10^7 \text{ s}/\text{m}^2$. We assume $R = 0.1 \text{ m}$ for the radius of the bread and find the time to reach thermal equilibrium $\tau_{\text{equil}} \simeq (c\rho/k) R^2 = 4 \times 10^5 \text{ s}$, or about 100 h. That is, it would take several days for the temperature to become uniform throughout the bread.

Bread baking occurs on the time scale of an hour, much shorter than the time to reach equilibrium: $t_{\text{bake}} \ll \tau_{\text{equil}}$. This implies that bread sustains a finite temperature difference between the center and the crust during baking. The center of the bread is approximately at the temperature $T_{\text{in}} = 320 \text{ K}$ when the crust is at the temperature $T_{\text{out}} = 425 \text{ K}$.⁶ If we assume that dough is prepared at room temperature (295 K), we find the rate of temperature change at the center of the bread to be $\Delta T_{\text{in}} / \Delta t \simeq (320 \text{ K} - 295 \text{ K}) / (3600 \text{ s}) \simeq 10^{-2} \text{ K/s}$.

The rising temperature induces physico-chemical changes, in particular starch gelation and the expansion of air bubbles in dough. The latter produces the crumb structure of bread. An inclusion of such processes is quite challenging in an analytical treatment and requires a computational fluid dynamics (CFD) treatment.⁷ The reader is referred to the papers by Mondal and Datta^{8,9} and Purlis¹⁰ where a more exhaustive overview of baking in the food-engineering literature can be found.

Moisture is important for texture and other physico-chemical properties of bread. In a basic bread recipe, 420 g of water are added to roughly 850 g of solid material (mostly

flour).¹ Flour also contains moisture depending on the relative humidity of the ambient air, which makes it difficult to accurately determine the water content of dough. However, we estimate the mass fraction of water in the dough to be of the order of one-third. Indeed, a water content of 37.5% for white bread has been reported in the literature.¹¹ This figure is consistent with our estimate for the water content of dough; it shows in particular that only a small fraction of water evaporates during baking. We refer to this difference between the water content in dough and bread as the *excess* water. The excess water content is nonzero in dough, decreases during baking, and is zero in bread. Since the initial water content κ_0 depends on a whole range of conditions, we treat the initial excess water content of dough as the only adjustable parameter of our model and choose fractions between 0.5% and 5% of the total mass.

While the mass of the excess water is small, it determines the timescale of bread baking due to the enormous value of the latent heat of evaporation. The fractional excess water content κ diminishes gradually from the crust to the center during baking. A loaf of bread has the approximate shape of a cylinder with radius R and length L so that $\kappa = \kappa(r, t)$ is a continuous function of the radius and time. This model would yield a system of nonlinear partial differential equations for the dynamic variables.

We expect that $\kappa(r, t = \text{constant})$ has the shape of a sigmoid function with a sharp drop-off at the boundary $r = \xi$ separating baked bread from dough. We simplify the description and treat the excess water content as a two-state variable: it is nonzero in dough (undone phase), that is, $\kappa = \kappa_0$ for $0 < r < \xi$, and zero in bread (done phase), that is, $\kappa = 0$ for $\xi < r < R$; see Fig. 1. The time dependence of bread baking is determined by the move of the crossover region at $r = \xi$ towards the center. To our knowledge, such a two-state model for baking has not been discussed in the literature.

Heat flow in a wide range of geometries is treated authoritatively in standard references.^{12,13} Cylindrical coordinates (r, ϕ, z) have singular behavior near $r = 0$, which poses a challenge in finite-element methods.¹⁴ In an analytical treatment, the singular behavior can be circumvented by considering a minimal cutoff radius r_{min} and then taking the limit $r_{\text{min}} \rightarrow 0$.

As we will discuss below, heat transfer during baking introduces nonlinear terms in the dynamical equations. A complete treatment requires that the nonlinearity be

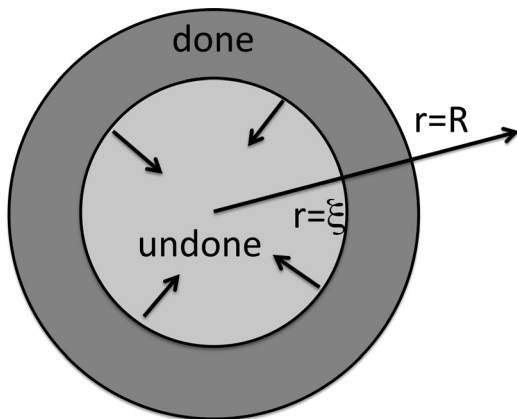


Fig. 1. The cross section of bread: the outer portion of the baked bread is done (bread), while the inner portion is undone (dough). The center of the bread is at the coordinate $r=0$, and the crust is at $r=R$. The crossover between the two phases is at the radius $r=\xi$.

combined with the singularity from the cylindrical coordinates. Such a comprehensive treatment would be appropriate for a paper in a specialized research journal. Our goal is less ambitious, in that we focus on the thermodynamics of baking that is determined by the principles of energy balance.

We consider the case when the cutoff radius $r_{\min}$ is small but nonzero, and ignore the decrease of the cylindrical surface area as the radius decreases. We therefore restrict ourselves to a one-dimensional geometry of a “loaf” of bread, as shown in Fig. 2, and derive a system of nonlinear *ordinary* differential equations in one spatial dimension (x).

We observe that the bread crust has a small but nonzero thickness and that furthermore baking is not a matter of seconds. These empirical facts suggest that we introduce a smallest length scale δx and shortest time scale δt in the mathematical description. In a coarse-grained description, the temperature gradient along the radius can have finite jumps, that is, $dT/dx|_{x^+} - dT/dx|_{x^-} > 0$.

The outline of the paper is as follows. In Sec. II, we present the simple two-state model for bread and describe the main processes involved in heat transport. In Sec. III, we derive the equations for the dynamical variables, and solve them numerically using the Runge-Kutta method. We summarize and discuss our main results in Sec. IV.

II. TWO-STATE MODEL OF BREAD

We consider a one-dimensional model for bread with $x=0$ at the center and $x=R$ at the surface, and choose $R=10$ cm as a typical radius of a loaf of bread. Bread has a thin crust of about a millimeter in thickness, which we associate with the length scale $\delta x = 1.0 \times 10^{-3}$ m. The corresponding time scale for heat conduction is $\delta t \simeq (c\rho/k)(\delta x)^2 \simeq 4 \times 10^7 \text{ s/m}^2 \cdot (10^{-3} \text{ m})^2 = 40$ s. Thus, a coarse-grained description is consistent with the common experience that baking is *not* a matter of seconds.

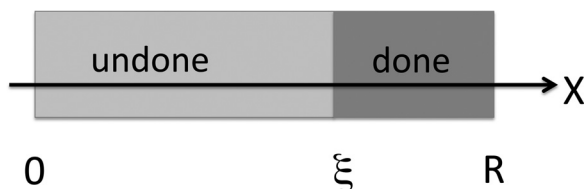


Fig. 2. The one-dimensional model of bread: the center of the bread is at $x=0$, and the crust is at $x=R$. The crossover between the two phases is at $x=\xi$.

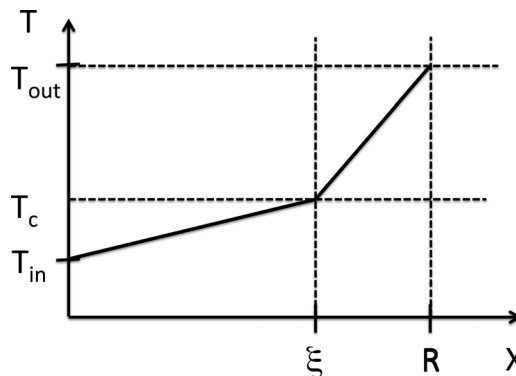


Fig. 3. Temperature profile $T(x)$ along the bread. The crust is at the surface temperature $T(R) = T_{\text{out}}$ and the center is at the temperature $T(0) = T_{\text{in}}$; both temperatures are monotonically increasing as a function of time. The crossover coordinate is at the fixed temperature $T(\xi) = T_c$.

We estimate the temperature gradient from the inside and outside temperature, $\nabla T \simeq (T_{\text{out}} - T_{\text{in}})/R = (425 \text{ K} - 320 \text{ K})/R = 1.0 \times 10^3 \text{ K/m}$. We conclude that the spatial coarse-graining is associated with a temperature variation $\delta T = \nabla T \cdot \delta x = 1.0 \text{ K}$; our estimate shows that the noticeable temperature increase is about one kelvin.

Since $j = -k\nabla T$, a discontinuity of the temperature profile is associated with a finite jump of the heat flux, and we expect a piecewise-linear dependence of the temperature profile, as shown in Fig. 3. A jump in the heat flux is due to a rise in the temperature δT of a layer with thickness δx during the time δt so that the necessary power is $\delta Q/\delta t = \rho A \delta x \delta T/\delta t$. We estimate $\delta T/\delta t$ from the temperature rise of the crust, $\delta T/\delta t \simeq \Delta T_{\text{out}}/t_{\text{bake}} = (450 \text{ K} - 320 \text{ K})/3200 \text{ s} = 0.05 \text{ K/s}$, and find $\delta j = \rho c (\Delta T_{\text{out}}/\Delta t) \delta x = [2000 \text{ J/(kg} \cdot \text{K)} \cdot 1000 \text{ kg/m}^3 \cdot 0.05 \text{ K/s}] / (1.0 \times 10^{-3} \text{ m}) = 100 \text{ W/m}^2$; this value is consistent with our estimate below for the overall heat flux in bread.

A standard oven temperature for bread baking is $T_{\text{oven}} = 450 \text{ K}$. We assume blackbody radiation with emissivity $e = 1$ so that the net heat influx is given by the Stefan-Boltzmann law,

$$j_{\text{net}} = \sigma T_{\text{oven}}^4 - \sigma T_{\text{out}}^4, \quad (1)$$

where $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \text{ K}^4)$. As the crust changes its temperature from its initial value $T_{\text{out}}(0) = 300 \text{ K}$ to the final value $T_{\text{out}}(t_{\text{bake}}) = 420 \text{ K}$, the net heat influx drops from $j_{\text{net}}(0) \simeq 1850 \text{ W/m}^2$ to $j_{\text{net}}(t_{\text{bake}}) \simeq 500 \text{ W/m}^2$. In Eq. (1), we ignore convective heat transport from the turbulent flow of air, which is important in real ovens.¹⁵

We choose the mass fraction of excess water κ as the state variable for bread baking. The fraction is nonzero in dough and is zero in bread:

$$\kappa = \begin{cases} \kappa_0 & \text{undone ("dough"),} \\ 0 & \text{done ("baked bread").} \end{cases} \quad (2)$$

That is, the bread is divided into distinct *done* (baked) and *undone* (unbaked) phases, separated at the coordinate $x = \xi$, as shown in Fig. 2. The separation of the two phases is characterized by a crossover temperature T_c , which we take to be 330 K. For times $t > 0$, the *done* phase moves toward the center so that $d\xi/dt < 0$. The baking time is then defined by

$$\xi(t_{\text{bake}}) = 0, \quad (3)$$

i.e., when the crossover region reaches the center.

The outside of the bread (the crust) is at the temperature T_{out} , and the center is at the temperature T_{in} . The temperatures at the two boundaries increase with time: $dT_{\text{out}}/dt > 0$ and $dT_{\text{in}}/dt > 0$. For the initial condition, we assume that the excess water has evaporated from the crust, and thus set the initial temperature of the crust equal to the crossover temperature, $T_{\text{out}}(0) = T_c$. We point out that this initial condition is consistent with the instruction in some recipes: bread is placed in a very hot oven for a short time until the oven temperature is set at a lower temperature so that the water inside a thin crust evaporates very quickly before the rest of the bread undergoes any change.

The net radiative heat flux entering the bread is $j_{\text{net}} = \sigma(T_{\text{oven}}^4 - T_{\text{out}}^4)$, and the heat flux leaving the crust is determined by the temperature gradient $k(T_{\text{out}} - T_c)/(R - \delta x - \xi)$. We assume that the crust has thickness δx and find the time dependence of the surface temperature,

$$\frac{dT_{\text{out}}}{dt} = \frac{1}{\rho c \delta x} \left[\sigma(T_{\text{oven}}^4 - T_{\text{out}}^4) - k \frac{T_{\text{out}} - T_c}{R - \delta x - \xi} \right]. \quad (4)$$

As the temperature T_{out} increases during baking, the net radiative heat influx decreases, and the overall baking process slows down.

When the crossover region moves toward the center from ξ to $\xi - d\xi$, heat is dissipated by the evaporation of the excess water and the change in the temperature profile along the bread. We estimate the temperature increase as $\delta T = (\Delta T / \Delta x) \delta \xi$, so that the required heat flux is $j_c = \rho[\kappa_0 L + c(\Delta T / \Delta x)R] d\xi/dt$, where $L = 33.5 \times 10^5 \text{ J/kg}$ is the latent heat of water. If the (average) temperature variation along the bread is $\Delta T \simeq 10 \text{ K}$ and $\kappa_0 \simeq 0.05$, we find the estimates $\kappa_0 L \simeq 0.05 \cdot 22.4 \times 10^5 \text{ J/kg} \simeq 1 \times 10^5 \text{ J/kg}$ and $c(\Delta T / \delta x)R \simeq 2 \times 10^3 \text{ J/(kg} \cdot \text{K)} \cdot 10 \text{ K} \simeq 2 \times 10^4 \text{ J/kg}$. We find $\kappa_0 L > c(\Delta T / \delta x)R$ so that the heat necessary to evaporate the excess water in the crossover region is much greater than the heat necessary to change the temperature profile in both dough and bread. It follows that the evaporation of excess water determines the overall time dependence of baking, by governing the velocity of the crossover region. We write

$$\frac{d\xi}{dt} = -\frac{k}{\kappa_0 L \rho} \left[\frac{T_{\text{out}} - T_c}{R - \delta x - \xi} - \frac{T_c - T_{\text{in}}}{\xi - \delta x} \right]. \quad (5)$$

We use κ_0 as the single free parameter in our model and choose a range that is consistent with approximate empirical values.

Heat flows from the crossover region to the center, which leads to a rise in the temperature of the center of the loaf, T_{in} . We assume the same spatial coarse graining δx for the crust and the center and obtain

$$\frac{dT_{\text{in}}}{dt} = \frac{k}{\rho c \delta x} \frac{T_c - T_{\text{in}}}{\xi - \delta x}. \quad (6)$$

In the proposed two-state approximation, bread baking is described by three dynamic variables: the temperatures of the crust and center of the bread, T_{out} and T_{in} , respectively, and the crossover coordinate ξ . Equations (4)–(6) contain *nonlinear* terms $(T_{\text{oven}}^4 - T_{\text{out}}^4)$, $(R - \delta x - \xi)^{-1}$, and $(\xi - \delta x)^{-1}$, which cannot be linearized without drastically changing the time dependence of the dynamic variables.

III. DYNAMIC EQUATIONS AND RESULTS

We use $\delta x = 1.0 \times 10^{-3} \text{ m}$ for the spatial coarse-graining, and $T_{\text{oven}} = 450 \text{ K}$ for the constant oven temperature. The crossover coordinate ξ between the done and undone phases is defined by $T(\xi) = T_c = 330 \text{ K}$. We consider bread with “radius” $R = 0.1 \text{ m}$. The center of the bread is initially at room temperature, $T_{\text{in}}(0) = 293 \text{ K}$. We ignore the short time interval during which the crust develops, and arrive at the initial conditions $\xi(0) = R - 2\delta x = 9.8 \times 10^{-2} \text{ m}$ and $T_{\text{out}}(0) = T_c$.

We next introduce a set of scaled dynamical variables and parameters. The fixed oven temperature defines a characteristic temperature, so we define

$$T' = \frac{T}{T_{\text{oven}}}, \quad (7)$$

so that $T' < 1$. The characteristic length scale is the “radius” of the bread R , so we use the scaled crossover coordinate

$$\xi' = \frac{\xi}{R}, \quad (8)$$

with $\xi' < 1$.

There is no intrinsic time scale for baking that is determined by the properties of dough alone because baking is a non-equilibrium process maintained by a finite temperature difference between the oven and bread. The baking time is determined by the oven temperature: we expect that a hotter oven temperature leads to a shorter time. We conclude that the time scale for baking must be determined self-consistently. For the oven temperature $T_{\text{oven}} = 450 \text{ K}$, we invoke baking experience and choose one hour as the appropriate scale, $\tau = 3.6 \times 10^3 \text{ s}$. We then define the scaled time,

$$t' = \frac{t}{\tau}. \quad (9)$$

The self-consistency of this choice will be confirmed if the calculation yields a scaled baking time of the order of unity, $t'_{\text{bake}} \sim \mathcal{O}(1)$. We use the term “appropriate” time scale rather “characteristic” time scale because we expect that τ depends on the oven temperature and the bread thickness.

We now re-express the dynamical equations (4)–(6) in scaled units. We use the typical values for the material constants and oven temperature given earlier, and round each computed combination to one or two significant figures so that, for example, $\sigma T_{\text{oven}}^4 \simeq 2 \times 10^3 \text{ J/(s} \cdot \text{m}^2)$ and $c\rho/k \simeq 4 \times 10^6 \text{ s/m}^2$. Then, in terms of the scaled variables (but with the primes now removed for simplicity), the dynamical equations become

$$\frac{dT_{\text{out}}}{dt} = 10(1 - T_{\text{out}}^4) - 8 \left(\frac{T_{\text{out}} - 0.72}{0.99 - \xi} \right), \quad (10)$$

$$\frac{T_{\text{in}}}{dt} = 8 \left(\frac{0.72 - T_{\text{in}}}{\xi - 0.01} \right), \quad (11)$$

$$\frac{d\xi}{dt} = -\frac{0.05}{\kappa_0} \left(\frac{T_{\text{out}} - 0.72}{0.99 - \xi} - \frac{0.72 - T_{\text{in}}}{\xi - 0.01} \right). \quad (12)$$

This is a set of three coupled, nonlinear differential equations for the three scaled variables T_{out} , T_{in} , and ξ . Our results will

depend only weakly on the values of the dimensionless prefactors 10, 8, and 0.05 obtained from the typical values for material constants given above (and rounded to one significant figure). This weak dependence is consistent with baking experience: for example, the baking time does not significantly change if eggs are added to the recipe, as in challah and sweet breads.

The right-hand sides of Eqs. (10) and (11) define scaled heat fluxes, which we write here for future reference. The heat flux from the oven into the bread is

$$j_{\text{net}} = 10(1 - T_{\text{out}}^4), \quad (13)$$

while the heat flux from the crust to crossover region is

$$j_+ = 8 \left(\frac{T_{\text{out}} - 0.72}{0.99 - \xi} \right), \quad (14)$$

and the heat flux from the crossover region to the center of the bread is

$$j_- = 8 \left(\frac{0.72 - T_{\text{in}}}{\xi - 0.01} \right). \quad (15)$$

We do not simplify the nonlinearity of this dynamical system; instead we solve the system numerically and present the results graphically. We use a Runge-Kutta routine¹⁶ built into the MathCad environment¹⁷ with initial conditions $T_{\text{in}}(0) = 0.65$, $T_{\text{out}}(0) = 0.72$, and $\xi(0) = 0.98$. We choose 1000 integration steps from the initial time $t=0$ to the final time t_f , chosen to be slightly longer than the baking time t_{bake} . Our results are not sensitive to the number of integration steps.

Figure 4 shows the time dependence of the crossover location, for four different values of the excess water fraction κ_0 . The speed of the crossover $v = |d\xi/dt|$ monotonically decreases with time. This slowdown is a consequence of the rising temperature of the crust and the consequent decrease in the net heat influx from the oven into the bread. The maximum speed is approximately three times greater than the minimum speed, $v_{\text{max}}/v_{\text{min}} \simeq 3$, as shown in Table I. The order of magnitude of the minimum speed is 1 (i.e., 1 mm/min) and corresponds to a baking time of

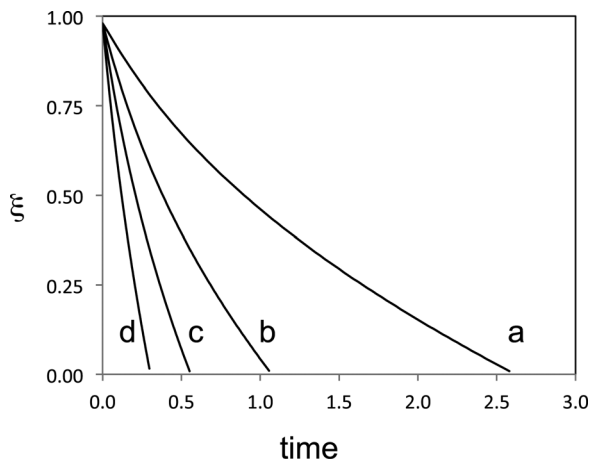


Fig. 4. The scaled coordinate ξ of the crossover region as a function of the scaled time t for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

Table I. The maximum and minimum speeds of the crossover region, in scaled and conventional units, for selected values of the excess water content in dough (κ_0).

κ_0	v_{max} (scaled)	v_{max} (cm/min)	v_{min} (scaled)	v_{min} (cm/min)
0.005	6.0	1.00	2.3	0.28
0.01	2.8	0.47	1.2	0.20
0.02	1.8	0.33	0.6	0.10
0.05	0.7	0.12	0.2	0.04

approximately 1 h. We note that the minimum speed is equal to the ratio of spatial and temporal coarse-graining: $v_{\text{min}} \simeq \delta x / \delta t$.

For the assumed excess water content, $5 \times 10^{-3} < \kappa_0 < 5 \times 10^{-2}$, we find baking times of the order of unity in scaled units: $t_{\text{bake}} \sim \mathcal{O}(1)$. We emphasize that this is *not* a trivial consequence of our choice of units, but rather confirms that $\tau = 3.6 \times 10^3$ s is the appropriate time scale for baking bread with size $R = 0.1$ m at the specified oven temperature. The baking times as a function of the excess water content are given in Table II and shown in Fig. 5. The results are well fit by a straight line, which goes through the origin because the baking time approaches zero for $\kappa_0 = 0$:

$$t_{\text{bake}} = 52 \kappa_0, \quad (16)$$

in dimensionless units (hours of time). From experience, a typical baking time for bread is approximately 50–60 min; this would correspond to 2% excess water content in the dough, which is a reasonable value.

The time dependence of the crust temperature is shown in Fig. 6. The temperature increases rapidly during an initial time period and then increases only gradually during the remainder of baking time.

The temperature at the end of baking, $T_{\text{out}}(t_{\text{bake}}) = 0.94$, is independent of the excess water content κ_0 :

$$T_{\text{out}}(t_{\text{bake}}) = 0.94, \quad (17)$$

or 425 K in conventional units. We now show that this result can be obtained by linearizing Eq. (10). We write $T_{\text{out}} = 1 - \tilde{T}$ and set $\xi \simeq 0$ since the crossover region is near the center at the end of baking. From Eq. (10), we find that $d\tilde{T}/dt \simeq -(48\tau - 2.24)$. We thus see that $d\tilde{T}/dt = 0$ at $\tilde{T}_f = 2.24/48 = 0.05$, or $T_{\text{out},f} = 0.95$. In physical units, the corresponding expression reads

$$T_{\text{oven}} - T_{\text{out}}(t_{\text{bake}}) \simeq \frac{T_{\text{oven}} - T_c}{1 + 4R/\lambda}, \quad (18)$$

where we introduce the length scale

Table II. Baking time, in scaled and conventional units, for selected values of the excess water content in dough (κ_0).

κ_0	t_{bake} (scaled)	t_{bake} (minutes)
0.005	0.27	16
0.01	0.54	32
0.02	1.05	63
0.05	2.58	155

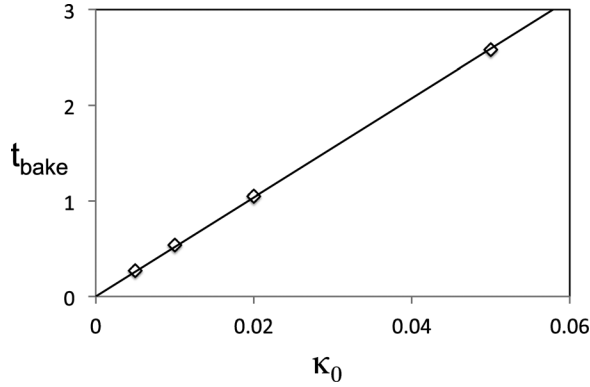


Fig. 5. Baking time (in scaled units) as a function of the excess water content of dough. The straight-line fit is $t_{\text{bake}} = 52 \kappa_0$.

$$\lambda = \frac{k}{\sigma T_{\text{oven}}^3}. \quad (19)$$

We find the numerical value $\lambda \simeq 0.1$ m, which is approximately equal to the radius of bread, so that $\tilde{T} \simeq (T_{\text{oven}} - T_c)/5 \simeq (450 \text{ K} - 325 \text{ K})/5 = 25 \text{ K}$. Thus, we recover the above result for the crust temperature, $T_{\text{out}}(t_{\text{bake}}) \simeq 425 \text{ K}$, consistent with empirical values.

The relatively slow increase of the crust temperature toward the end of baking is due to the quartic temperature dependence of the net heat influx [see Eq. (13)]. The initial heat influx is $j_{\text{net}}(0) = 7.3$ (1700 W/m² in conventional units), which monotonically drops to the final value $j_{\text{net}}(t_{\text{bake}}) = 2.2$ (500 W/m² in conventional units). The time dependence of the net heat influx is shown in Fig. 7.

We now turn our attention to the behavior of the crust temperature as a function of the excess water content. The initial growth rate of the surface temperature $dT_{\text{out}}/dt|_0$ shows a four-fold increase when the excess water content drops by a factor of 1/10, as shown in Table III.

The heat current from the crust to the crossover region is shown in Fig. 8. This current exhibits non-monotonic behavior: there is a rapid increase from the initial value $j_+(0) = 0$ to a maximum value $j_{+, \text{max}} = j_+(t_{\text{max}})$ at time $t_{\text{max}} \simeq 0.005$ (about 20 s). Because this time is shorter than the temporal coarse-graining $\delta t \simeq 40$ s, the initial rapid increase is not shown in Fig. 8. The maximum value of the current depends weakly on the excess water content, as shown in Table IV,

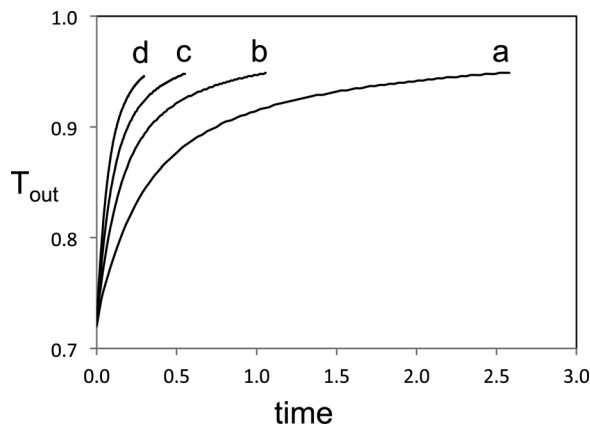


Fig. 6. Scaled temperature of the crust as a function of the scaled time for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

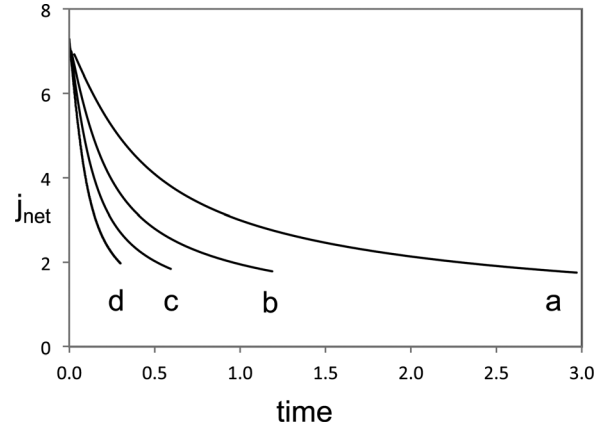


Fig. 7. Time dependence (in scaled units) of the net heat influx through the crust $j_{\text{net}}(t)$ [Eq. (13)] for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

and is approximately $j_{+, \text{max}} \simeq 6$ (1500 W/m² in conventional units). The flux j_+ drops to the final value $j_+(t_{\text{bake}}) \simeq 1.8$ (400 W/m² in conventional units) at the end of baking, independent of the excess water content.

The temperature at the center changes from the initial $T_{\text{in}}(0) = 0.65$ (295 K) to a final value $T_{\text{in}, f} = T_c = 0.72$ (325 K), as shown in Fig. 9. The initial rate of increase is

$$\left. \frac{dT_{\text{in}}}{dt} \right|_0 = 0.52 \quad (3.6 \text{ K/min}), \quad (20)$$

independent of the fraction of excess water κ_0 . The value $dT_{\text{in}}/dt|_0 = 3.6 \text{ K/min} = 6 \times 10^{-2} \text{ K/s}$ is about six times greater than the empirical average value over the entire baking time, $\Delta T_{\text{in}}/\Delta t \simeq 10^{-2} \text{ K/s}$. The difference between initial and average values reflects the decrease in heat transport through the bread during baking.

The temperature at the center of the bread rises more slowly than the temperature of the crust: $dT_{\text{in}}/dt|_0 < dT_{\text{out}}/dt|_0$, as can be seen by comparing Eq. (20) to the values in Table III. The rise of the temperature at the center is approximately $\Delta T_{\text{in}} \simeq 0.72 - 0.65 = 0.07$ in scaled units, as shown in Fig. 9, so that a value $dT_{\text{in}}/dt|_0 \simeq 0.52$ gives the crossover time $t_{\text{in}, c}$ after which the temperature rise at the center is negligible:

$$t_{\text{in}, c} = \frac{0.07}{0.52} = 0.14 \quad (8 \text{ min}). \quad (21)$$

Since the baking time is on the order of one hour, we have $t_{\text{in}, c} \ll t_{\text{bake}}$ and we conclude that the temperature at the center is not a useful indicator for determining whether bread is done.

The asymptotic value $T_{\text{in}} = 0.72$ is a consequence of the expression for the heat flux from the crossover region to the

Table III. The initial rate of increase of the surface temperature, in scaled and conventional units, for selected values of the excess water content in dough (κ_0).

κ_0	$dT_{\text{out}}/dt _0$ (scaled)	$dT_{\text{out}}/dt _0$ (K/min)
0.005	2.62	20.0
0.01	1.70	12.5
0.02	1.17	9.0
0.05	0.60	4.8

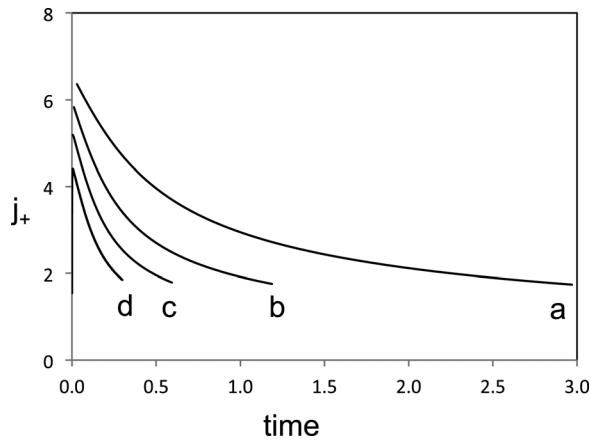


Fig. 8. Time dependence (in scaled units) of the heat flux j_+ from the crust to the crossover region [Eq. (14)] for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

center: $j_- = 8(0.72 - T_{in})/(\xi - 0.01)$ in our model. The time dependence of j_- is shown in Fig. 10. Similar to our discussion of j_+ , we ignore the rapid rise during a very short initial time interval Δt . The initial value is

$$j_-(0) \simeq 0.6 \quad (130 \text{ W/m}^2), \quad (22)$$

and $j_-(t)$ rapidly approaches zero for $t > t_{in,c} \simeq 0.2$ (12 min). While the dependence on excess water content is quite weak, it is nontrivial: for times $t < t_{in,c}$, the flux is greater for larger water content, whereas for times $t > t_{in,c}$ the heat flux is smaller for larger water content.

IV. SUMMARY AND DISCUSSION

Problem solving is a key expected learning outcome for students in introductory physics, and the usefulness of *context-rich* problems is well documented.¹⁸ The situation in upper-level undergraduate courses is similar, and students greatly benefit when they are exposed to problems from an engineering point of view. The necessary steps include (i) identifying key physical principles, (ii) making proper assumptions, (iii) developing an appropriate mathematical model, (iv) undertaking parametric study of the mathematical model, and (v) interpreting the results.

Our goal in this paper is a relatively nontechnical (simplified) description of the thermodynamics of bread baking, and no attempt is made to make a detailed comparison with results from experiments or from more sophisticated computations. This paper illustrates how bread baking can be used to further develop critical thinking skills in a thermal physics course. We note that “engineering design” is included in the Next Generation Science Standards (NGSS).¹⁹ We discuss

Table IV. The maximum heat flux from the crust to the crossover region, in scaled and conventional units, for selected values of the excess water content in dough (κ_0).

κ_0	$j_{+,max}$ (scaled)	$j_{+,max}$ (W/m ²)
0.005	4.4	1025
0.01	5.2	1210
0.02	5.8	1350
0.05	6.4	1490

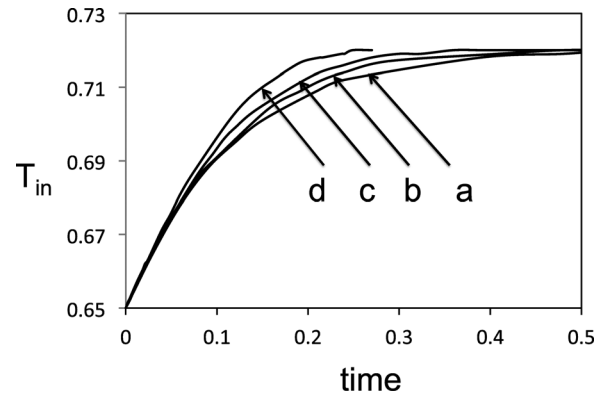


Fig. 9. Time dependence (in scaled units) of the temperature T_{in} at the center of the bread for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

the energy flow in bread (dough) during baking, and use an order-of-magnitude estimate to show that the system is in a nonequilibrium state.

A nonzero heat flux maintains a finite temperature gradient during the baking process. The rate-limiting step of the dynamic is the evaporation of excess water, that is, the difference of the water content between dough and bread. We simplify the spatial dependence of the excess water and propose a two-state model in which the mass fraction of excess water is a fixed constant in “dough” and zero in “bread.” The importance of the excess water content is supported by James Beard’s doneness test: “bake ... until the bread sounds hollow when tapped with the knuckles.”²⁰ Bread sounds hollow when the excess water has evaporated from the center.

Within this approximation, the dynamical equations for baking are coupled, nonlinear ordinary differential equations for the crossover coordinate ξ and the crust and center temperatures, T_{out} and T_{in} . The heat flux inside the bread is piecewise-constant with a discontinuity at ξ , which corresponds to a piecewise-linear temperature profile. The discontinuity in the heat flux is determined by the evaporation of excess water in a small crossover region.

If we allowed the excess water content to vary continuously, we would arrive at a system of coupled, nonlinear partial differential equations. The numerical solution would require the use of finite-element methods,²¹ which is more challenging than the straightforward Runge-Kutta integration

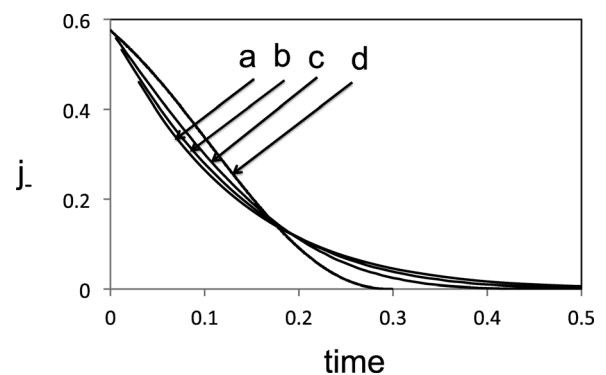


Fig. 10. Time dependence (in scaled units) of the heat flux j_- from the crossover region to the center of the bread [Eq. (15)] for (a) $\kappa_0 = 0.05$, (b) $\kappa_0 = 0.02$, (c) $\kappa_0 = 0.01$, and (d) $\kappa_0 = 0.005$.

used here. We expect that the temperature profile would be a convex function and follow an approximately quadratic behavior.²²

Our results are consistent with practical experience: it takes about one hour to bake a loaf of bread with radius $R = 0.1$ m in an oven of temperature $T_{\text{oven}} = 450$ K. Furthermore, we find that the surface temperature can be used to determine whether the bread is “done,” whereas the temperature at the center is an unreliable indicator for doneness.

We find that bread baking is associated with the length scale $\lambda = k/\sigma T_{\text{oven}}^3 \simeq 0.1$ m for typical values. This result shows that bread cannot be made arbitrarily large in diameter: if $R \gg \lambda$, the surface temperature decreases, while the necessary time for baking increases. Indeed, the earliest breads by Egyptians and Sumerians were thin (“flatbread”),²³ which likely reflects the difficulty in maintaining a high oven temperature for a sufficiently long period of time.

Our treatment ignores several important physico-chemical processes. A more realistic model of baking would include properties of small air pockets inside the bread. Since air is a good (thermal) insulator, this would decrease the coefficient of thermal conductivity k . Our one-dimensional model ignores the decrease of the boundary surface as it moves inward toward the center of the loaf. The correct geometry would increase the heat flux near the center, and consequently speed up baking towards the end. The inclusion of these effects is, however, beyond the intended scope of this paper.

ACKNOWLEDGMENTS

The author wishes to thank Ray Adamic, Ph.D. candidate in Mechanical Engineering, Cleveland State University, who introduced me to the science and engineering of bread baking. The author acknowledges the constructive criticism by three anonymous referees of this journal, which greatly helped to improve the manuscript. The author also wishes to thank Professor Majid Rashidi, Department of Engineering Technology, Cleveland State University, for help in the preparation of the final version of this manuscript.

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Leeds & Northrup Potentiometric Indicator

This small, portable potentiometer was made by Leeds & Northrup of Philadelphia in 1931, and cost approximately \$150. It was used primarily with thermocouple junctions to measure temperatures. L&N put the instructions inside the cover of its portable instruments so it was always available. The company used oak for the cases of portable instruments, and used splined corner joints for strength. The instrument is owned by Leonard Fashoway. (Photograph by Leonard Fashoway and notes by Thomas B. Greenslade, Jr., Kenyon College)